For examiners' use only

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2	
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Total	

EBU714U A

Joint Programme Examinations 2016/17

EBU714U Security and Authentication

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

NOT allowed: electronic dictionaries.

INSTRUCTIONS

1. **You must not take answer books, used or unused, from the examination room.**
2. Write only in black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Yasir Alfadhl, Dr Michael Chai

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Filename: 1617_EBU714U_A No answerbook required

Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

a) Answer the following questions: **[7 marks]**

   i) Within the context of security requirements, define **confidentiality** and explain which security mechanisms can achieve it. **(2 marks)**

   ii) Consider the scenario of an e-commerce website like *Taobao* where customers can register an account and purchase items online with secure transactions. Give examples of the security services the system should provide in terms of confidentiality, integrity, access control, availability and non-repudiation. **(5 marks)**

Do not write in this column

7 marks

b) Answer the questions below about *one-time pad:* **[12 marks]**

i) One-time pad is a stream cipher. What is a stream cipher? **(2 marks)**

ii) A one-time pad uses the XOR function (⊕) as an encryption and decryption method. Given a plaintext of 11010 and key stream of 01110, what's the ciphertext? **(2 marks)**

iii) What is a brute-force attack? Is one-time pad vulnerable to brute-force attack? Explain your answer. **(8 marks)**

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		12 marks

c) DES is a block-cipher and its basic mode of operation is Electronic Code Book (ECB). In this mode of operation, if the same 64-bit block of plaintext appears more than once in a message, it always produces the same ciphertext. In order to avoid such a security weakness, DES can be deployed in different modes of operation such as the Cipher feedback (CFB) mode. Briefly explain how CFB mode works, with the help of diagrams. **[6 marks]**

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		6 marks

Question marking: $\frac{\quad}{7}+\frac{\quad}{12}+\frac{\quad}{6}=\frac{\quad}{25}$

Question 2

a) Using block diagrams, explain how would SHA1 process any length of data block to produce a unique 160-bits output. **[5 marks]**

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		5 marks

b) Explain the mechanisms of each of the following attacks: **[8 marks]**

i) ‘Replay’ attack **(1 mark)**

ii) 'Man-in-the-middle' attack **(7 marks)**

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		8 marks

c) Key distribution is one of the major problems when implementing a security mechanism. List two Public-key encryption algorithms which can be used to solve this problem. Explain how each of these methods can be used to exchange keys between two parties. **[12 marks]**

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		12 marks

Question marking: $\frac{\quad}{5}+\frac{\quad}{8}+\frac{\quad}{12}=\frac{\quad}{25}$

Question 3

a) What is a public key certificate? Explain what information a certificate contains to guarantee its integrity. **[5 marks]**

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	5 marks

b) A certificate includes a period of validity. Sometimes the certificate is revoked before it expires. **[5 marks]**

i) Give three reasons why a certificate should be revoked before its expiry date?

ii) How can Certification Authorities (CAs) maintain an up-to-date validity of all users and avoid invalid keys?

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	5 marks

c) Briefly explain how the Tunnel and Transport Modes operate in IPSec. **[4 marks]**

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d) In IPSec, what are the specific key exchange algorithms in ISAKMP? What are the roles of the Oakley key determination protocol and ISAKMP? **[5 marks]**

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		5 marks

e) List two techniques used by firewalls to control access and enforce a security policy. Explain each of them. **[6 marks]**

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	6 marks

Question marking: $\frac{\quad}{5}+\frac{\quad}{5}+\frac{\quad}{4}+\frac{\quad}{5}+\frac{\quad}{6}=\frac{\quad}{25}$

Question 4

a) Define the following two important SSL concepts: **[4 marks]**
   i) SSL connection **(2 marks)**
   ii) SSL session **(2 marks)**

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		4 marks

b) In PGP key management, there is a possibility of multiple public keys per user, therefore the recipient of the message needs to know which of his/her public keys was used for encryption. Explain how PGP identifies the public key. **[4 marks]**

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		4 marks

c) Draw a schematic diagram with adequate notations to show how a set of Private (PR_b) and Public (PU_b) keys of User B is used in the PGP's confidentiality operation. **[12 marks]**

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12 marks

d) Explain the Certificates Processing of S/MIME. **[5 marks]**

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		5 marks

Question marking: $\frac{}{4}+\frac{}{4}+\frac{}{12}+\frac{}{5}=\frac{}{25}$

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